COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 5

5 Computer Networking (awm22)

Foundations(1) and Networks(3): ICMP, topology, whole-system debugging, ping, traceroute

(*a*) A user complains that their web application `Times Out` after successfully connecting to a remote host, yet, an investigation with `ping` indicates the remote host is alive.

Propose a cause of this fault and outline a test-strategy that could be used automatically to detect such a fault. [4 marks]

Answer: A genuine risk of the candidate heading into the weeds in their answer....

A reasonable conjecture for the problem is the server based application has crashed, ping is an ICMP functionality dealt with in kernel so it works if the kernel works.

The application failure may be simply tested with a dummy, or test, or trivial operation (retrieving a known file).

Rule of thumb - want to test a system is working automatically? test it doing something "production-like" as possible....

Not likely to be a DNS (ping works) Not likely to be firewall (ping works) not likely to be routing (ping works) Successful connection implies not TCP itself

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(*b*) A user would like to infer the path between two hosts; they suggest using the utility `ping`. Make your argument for why `traceroute` is the more appropriate utility, comparing and contrasting their operation. [4 marks]

Answer: `ping` is a liveness test returning little more information than the RTT and the existence (via ICMP echo) of the remote interface.

`traceroute` is more cunning.

by default

traceroute sends packets to thee destination host for a (hopefully) unused UDP port. artificially deflated TTLs cause premature packet expiration, this in turn returns ICMP messages to the originator.

Triggering a sequences of such messages permits a path to be guessed.

naive interpretation of traceroute and ping output may infer/assume symmetric paths

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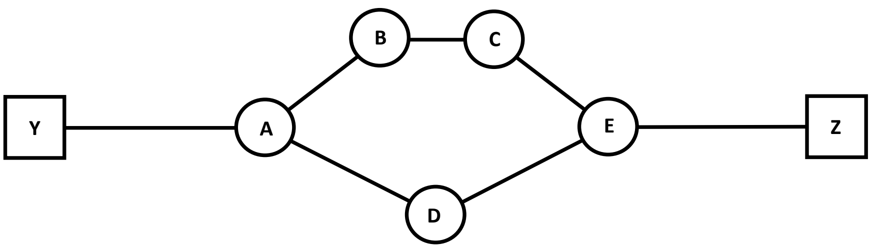
(*c*) Consider the test network below; Y and Z are two unix IPv4 hosts, while nodes A through E represent fully-conformant IPv4 routers providing the only connectivity between the two hosts.



Per-packet load balancing by A and E means packets may be sent on any valid path. No firewalls or packet filtering is used anywhere in the network.

(*i*) Using `traceroute` infers the following paths:
(Y-A-B-C-E-Z), (Y-A-D-E-Z), (Y-A-B-D-E-Z), and (Y-A-B-E-E-Z)

Explain (with diagrams as appropriate) why these results have occurred and identify the two true and two false results; [8 marks]

Answer: A long standing syndrome of traceroute, each test packet is independent of each other test. Thus a TTL =2 could elicit a response from B and C and a TTL = 3 could result in a response from D and E

(*ii*) A firmware upgrade for routers A and E means they now do per-flow and not per-packet load balancing. Despite the functions of the flow-based load balancing, `traceroute` results have not changed; two false paths are inferred along with two true paths.

Explain this outcome and a strategy to identify only true paths. [4 marks]

Answer: A few pairs of answers are possible.

One reasonable conjecture is that ICMP or UDP is not following the same rules as the flow-based LB. a solution is to use traceroute with synthetic TCP packets.

Possibly mentioning evaluating with both UDP and ICMP first.

A more sophisticated answer is to realise that traceroute uses progressive destination ports - each tuple will typically trigger a different hash for the flow-path.

One approach is to use packets that will select port numbers triggering the same flow-path. In reality this is quite tricky (we don't know the path pool or even the hash function); alternatively we could re-use the same destination port over and over.

Reusing the same port over and over runs the non-zero risk of triggering and potentially ignoring spurious duplicate responses in the network.
